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APPLIED RESEARCH

# Micromanipulation System Capable of Simultaneously Presenting High-Resolution and Large Field-of-View Images in Real-Time

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**ABSTRACT** Microinjection technology is widely used in biotechnological processes such as gene manipulation and microinsemination. Generally, microinjection is performed under an optical microscope while viewing the video of the targets in real-time. In the microinjection process, multiple oocytes must be placed in the same droplet, for which the injection must be performed multiple times. Furthermore, the target must be observed at different magnifications during injection and the oocyte transfer operation. Hence, the operators are required to change the magnification and light intensity repeatedly. These tasks are complicated and are a burden for the operators. Thus, the accuracy, reproducibility, and productivity of microinjection depend on the operator's skill level. In this study, to reduce the burden on the operator and simplify the operation of microinjection, we propose a micromanipulation system that enables both wide-range and high-resolution video presentation with free viewpoint selection. The proposed micromanipulation system is based on a view-expansion microscope system with simultaneous multi-view imaging using a galvanometer mirror and a high-speed vision system. We verify the effectiveness of the proposed system by evaluating the resolution of images presented to the operator and through the experiments involving moving microbeads performed by inexperienced subjects. The experimental results indicate that our micromanipulation system streamlines the task of micromanipulation by eliminating the need to change the objective lens.

**INDEX TERMS** Micromanipulation system, view-expansion microscope, macro-micro interaction.

## I. INTRODUCTION

Microinjection technology is widely used in gene manipulation and microinsemination. It has been used to create transgenic mice injecting foreign genes into pronuclear fertilized eggs of mice [1] and to create knockout mice that lack a specific endogenous gene [2]. Genetically modified animals like knockout mice are used not only for functional analysis of genes, but also for pathological analysis and therapeutic drug development as disease model animals [3], [4]. In addition, microfertilization using microinjection technique has significantly contributed to human infertility treatment [5].

When performing genetic manipulation on oocytes, the oocytes must be moved from the follicular fluid (or medium containing the oocytes collected from the follicle) to another dish several times as a preparatory step for microinjection. Moreover, detailed observation is required to find the oocytes from among the impurities present in the medium and to select those with a good cumulus cell status. This preparatory work requires wide field-of-view (FOV) microscopic

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observation at a high degree of magnification. A large number of oocytes are manipulated when performing microinjection on oocytes. Thus, the work space is divided into three spaces to avoid mixing the oocytes before and after the injection (Fig. 1). Oocytes before injection and oocytes after injection are placed in workspaces 1 and 3, respectively, and the injection work is performed in workspace 2. A large FOV is necessary when oocytes are moved from workspace 1 to 2 and from 2 to 3. In contrast, the injection work done in workspace 2 requires high resolution (high-magnification) imaging to perform dexterous work while observing details such as the polar body of the cell and the position of the nucleus. Conventionally, the wide-range images and high-resolution (high-magnification) images are obtained by manually changing the magnification of the objective lens and adjusting the light intensity corresponding to the objective lens. However, these tasks are cumbersome for the operator when performing microinjection on a large number of oocytes. Consequently, the accuracy, reproducibility, and productivity of microinjection depend on the operator's skill and fatigue levels.

A view-expansion microscope that achieves both large FOV imaging and high-resolution imaging by simultaneous multi-view video recording using a galvanometer mirror and high-speed vision has previously been proposed [6], [7]. However, the microscope system reported in these studies for microinjection is difficult to use. Two factors complicate its use: the defocusing caused by the change of the optical path length depending on the angle of the mirror and the difficulty in oocyte observation due to the use of epi-illumination. Based on the concept of the view-expansion microscope, we propose the use of transmitted illumination to image the manipulation of targets placed on a plane stage. In addition, the proposed microscope system comprises an electrically tunable lens (ETL) that can adjust its focal position to prevent defocusing caused by the change of optical path length.

Furthermore, we propose a micromanipulation system that enables both wide-range and high-resolution video recording with free viewpoint selection, which simplifies the microinjection process and reduces the burden on the operator. Algorithms to acquire and display both large FOV and high-resolution images are implemented in the proposed system. The effectiveness of the proposed system is verified by evaluating the resolution of images presented to the operator and through the experiments on moving microbeads. The initial contribution of this research was partly presented at IEEE International Conference on Robotics and Automation in 2021 [8]. Improvements to this paper include the design of the optical system using an electrically tunable lens, evaluation of the proposed system using a test target, and evaluation of the proposed system through experiments performed by inexperienced subjects.

The remainder of this paper is organized as follows. Section II introduces previous research work related to this study. Section III presents an outline of the proposed micromanipulation system and its specifications. Section IV describes the method to control the focal position of the system using the ETL. Section V explains the implemented algorithms to acquire and present both large FOV and high-resolution images. Section VI shows the evaluation experiments of the proposed micromanipulation system. The evaluation experiments include resolution evaluation using test target and evaluation of task time in micromanipulation. Section VII presents the conclusions of the study and discusses its future scope.

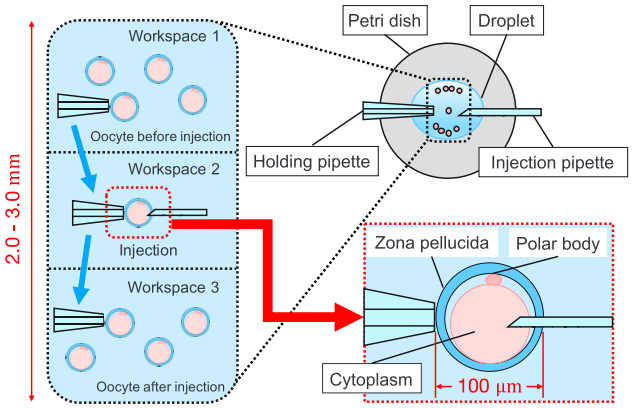



**FIGURE 1.** Microinjection process.

## II. RELATED WORKS

In this study, we built a micromanipulation system based on a microscope that is capable of both large field-of view imaging and high-resolution imaging. Conventional microscopy systems are subject to a trade-off between FOV and resolution; thus, several researchers have been trying to develop microscope systems that can achieve both large FOV and high-resolution imaging.

In certain microscopes, the microscope stage is moved and images are captured from multiple viewpoints to expand the FOV without compromising on resolution [9]. However, the microscope stage cannot be easily moved at high speed to follow the microinjection targets. Hence, this system is not suitable for microinjection as it requires real-time imaging. In another study, a microscope enabling high-speed viewpoint selection was developed by inserting a dual-axis mirror between the objective and imaging lens [10]. In addition, a deformable mirror is used to prevent aberration in the light passing through the outer edge of the objective lens and maintain high-resolution imaging. However, this method is not versatile as it requires the exclusive use of a huge objective lens and a deformable mirror. Another study reported a two-axis microscope with two separate scan paths and two multi joint arms that expands the FOV while maintaining resolution [11]. However, as the observable position is fixed, capturing physically continuous positions is difficult. Another microscope system was developed by inserting a novel optical system in the space between the objective lens and the object [12]. This system captures microscopic images from different viewpoints every 10 ms. However, the movable

range of the viewpoint is limited, and the FOV is insufficient for micromanipulation.

Furthermore, digital holographic microscopes (DHMs) that are capable of wide area imaging have been widely researched. Inline holographic microscopes require multiple shots to increase the resolution; thus, they are unsuitable for micromanipulation, which requires visual information in real-time [13], [14]. Phase-shifting holographic microscopes are also unsuitable for micromanipulation because they require multiple shots, and it takes time to collect images. To speed up the process, off-axis DHMs [15], [16] and parallel phase shift DHMs [17] were developed. These obtain high-resolution three-dimensional information from one-shot photography that shortens image acquisition time.

DHMs that attempt to display reconstructed images in real-time through high-speed image processing using GPU or CPU have also been developed. High-speed tomography systems that can measure and visualize $96^3$ voxels in 1.3 seconds were proposed using high-speed image acquisition and parallel processing using GPU [18], [19]. These systems were further improved, and a 3D reconstruction rate higher than the video rate of 25 frames/s (fps) has been achieved with a 256×256 pixel interferogram that has 73 different projection angles [20]. A holographic approach using interferometry that doubles the FOV of a holographic microscope without reducing the acquisition frame rate has also been reported [21]. A real-time DHM capable of observing a range of 4.8 ×4.8 $mm^2$ with a resolution of 11 $\mu$m [22] has been developed; another with a video rate over 35 fps to display a 2048×2048 hologram has been achieved using CPU and GPU [23], [24]. These high-speed DHMs have the advantage of recording three-dimensional information of target objects in a single shot. However, when the FOV is increased, the image resolution decreases and the time required for image reconstruction increases. Therefore, DHMs are unsuitable for microinjection, which requires a resolution of a few micrometers, visual field range of 3×3 mm, and visual feedback in real-time.

As described above, micromanipulation systems that display microscopic images in real-time and have both a large FOV and high-resolution have not been developed. Our research group has previously developed view-expansion microscopes that can acquire both large FOV and high-resolution videos using high-speed image acquisition and high-speed view-point control with a galvanometer mirror [6], [7]. The view-expansion microscopes developed in these studies are microscope systems that can freely change the viewpoint, FOV, and resolution. Moreover, the image acquisition function of these microscope systems is suitable for microinjection. However, they cannot be directly applied to microinjection owing to the inability to make the microscope stage flat and the use of epi-illumination. In this study, based on the concept view-expansion microscopes, we develop an inverted microscope system with transmitted illumination that can observe a target on a flat stage and can be applied to microinjection. Finally, we develop a micromanipulation system that enables both large FOV and high-resolution imaging and allows free viewpoint selection.

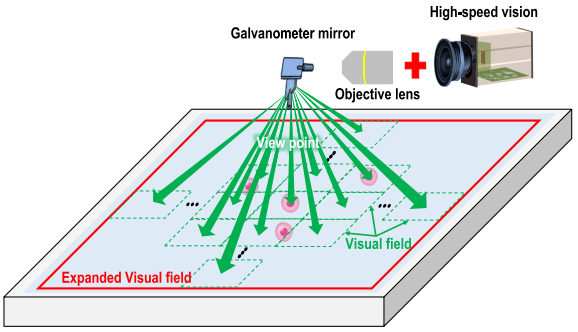



**FIGURE 2.** Concept of view expansion microscope.

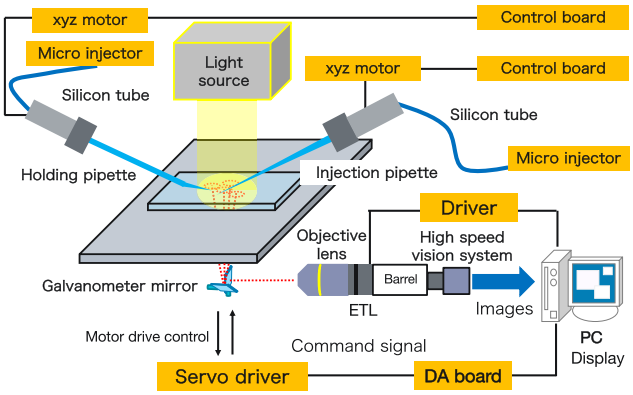



**FIGURE 3.** Configuration of the proposed micromanipulation system.

## III. PROPOSED MICROMANIPULATION SYSTEM

The proposed micromanipulation system is based on a view-expansion microscope that obtains focused images of the target placed on the flat microscopic stage from various viewpoints. The microscope system is realized by utilizing an ETL that provides focus position adjustment to expand the FOV. The resolution is simultaneously maintained using multi-view imaging with a galvanometer mirror and high-speed vision. Fig. 2 shows the conceptual schematic of the novel view-expansion microscope system.

In this study, we propose a micromanipulation system developed by adding micromanipulators to the novel view-expansion microscope system. Fig. 3 and Fig. 4 show the system configuration and overview of the proposed micromanipulation system, respectively. The micromanipulation system can adjust the viewpoint and FOV with a single camera-lens system based on simultaneous multi-view imaging using a galvanometer mirror and high-speed vision [25]. The system comprises an inverted microscope (IX73, OLYMPUS), two microinjectors (FemtoJet 4i, CellTram 4r Air, Eppendorf), two micro manipulators (TransferMan4r, Eppendorf), microscope unit (KTL-N21B-1, Kyowa Optics),

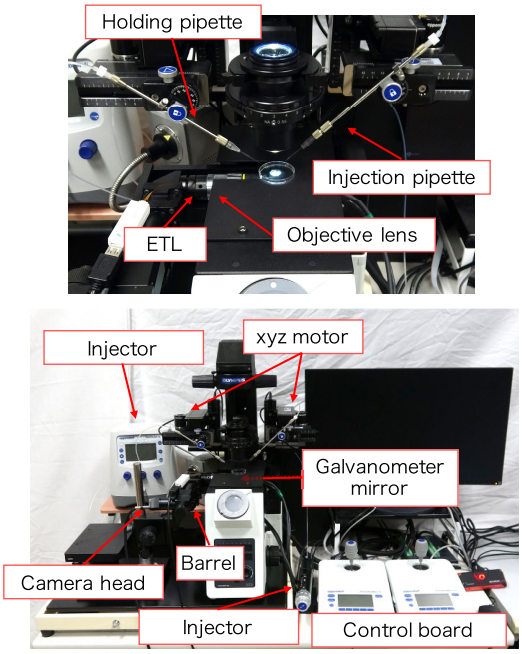



**FIGURE 4.** Overview of the proposed micromanipulation system.

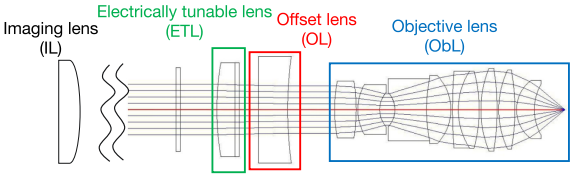



**FIGURE 5.** Optical model of the proposed system.

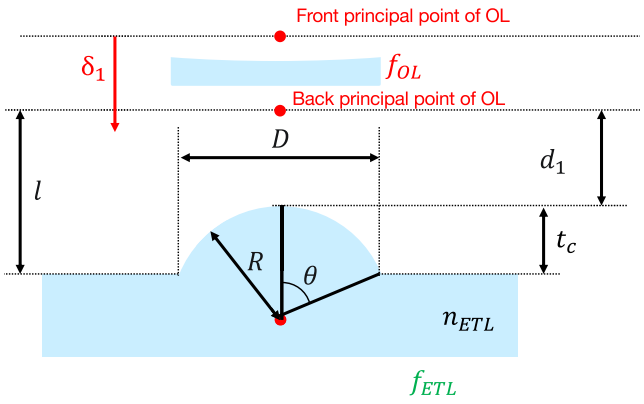



**FIGURE 6.** Focal distance and distance between ETL and offset lens.

objective lens (LWD95mm 10X, Kyowa Optics), galvanometer mirror with two degrees of freedom (6210 HSM 6mm, Cambridge Technology), high-speed vision (MQ003CG-CM, Ximea), ETL (EL 10-30-C-VIS-LD-MV, Optotune), control PC (Windows 10 Home 64bit OS, DESKTOP-U04CFTQ, Intel Core(TM) i9-9900KF CPU 3.60GHz, DDR32GB memory), D/A board (PCX-340416, Interface), an illuminator (LA-HDF158AS, HAYASHI WATCH-WORKS CO., LTD), and a liquid-crystal display (FlexScan EV2781, EIZO). The objective lens has a magnification of 10× and a working distance (WD) of 33 mm. The high-speed vision can capture 640×480 pixel images at a maximum rate of 500 fps. The pixel pitch of the high-speed vision and the liquid-crystal display are 7.4 $\mu$m and 0.233 mm respectively. The real-scale magnification of this system is approximately 315 times. In addition, in-focus images can be obtained from all viewpoints using the focus position adjustment of the ETL. The focus adjustment corresponds to the change in optical path length that depends on the angle of the mirrors.

## IV. FOCAL POSITION ADJUSTMENT USING AN ELECTRICALLY TUNABLE LENS

### A. RELATIONSHIP BETWEEN SYSTEM OPTICS AND ETL FOCAL LENGTH

The optical system of the proposed micromanipulation system comprises an imaging lens, electrical tunable lens, offset lens, and objective lens as shown in Fig. 5. Fig. 6 shows a schematic of the ETL and the offset lens. $D$ is the diameter of the variable focus lens that can be deformed. $d_1$ is the distance from the rear principal point of the offset lens to the front principal point of the variable focus lens, $R$ is the radius of curvature of the variable focus lens, $n_{ETL}$ is the refractive index of the liquid in the variable focus lens, and $t_c$ is the distance from the principal point of the variable focus lens to the front surface of the variable focus lens. First, let us consider the focal length of the ETL $f_{ETL}$ and the distance between the variable focus and the offset lenses $d_1$. Considering that the ETL changes the focal length by deformation, $t_c$ can be derived as follows:

$$\begin{aligned} t_c &= R(1-\cos\theta) \\ &= R\left(1-\sqrt{1-\frac{D^2}{4R^2}}\right). \end{aligned} \tag{1}$$

The focal length of the variable focus lens is 100–200 mm, $D$ is 10 mm, and the refractive index $n_{ETL}$ is 1.3. Thus, $\frac{D^2}{4R^2}$ is considerably smaller than 1. Therefore, it can be approximated as follows:

$$t_c = \frac{D^2}{8R} = \frac{D^2}{8(n_{ETL}-1)f_{ETL}} = \frac{D_2}{f_{ETL}}, \tag{2}$$

$$D_2 = \frac{D^2}{8(n_{ETL}-1)}. \tag{3}$$

From Eqs. (2) and (3), $f_{ETL}$ and $d_1$ are derived as follows:

$$d_1 = l - \frac{D_2}{f_{ETL}}, \tag{4}$$

$$f_{ETL} = \frac{D_2}{l-d_1}. \tag{5}$$

Next, let us consider replacing the ETL, offset lens, and objective lens with a single lens as shown in Fig. 7. Then, the focal length and principal point position of the proposed

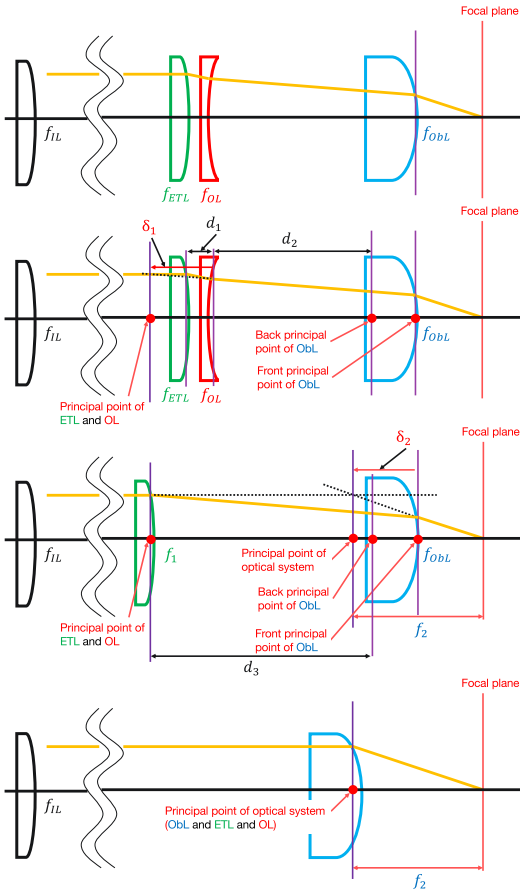



**FIGURE 7.** Calculation of the focal distance and the principal point of the optical system.

system can be derived as:

$$f_1 = \frac{f_{ETL} f_{OL}}{f_{ETL} + f_{OL} - d_1}, \tag{6}$$

$$\delta_1 = \frac{f_{OL} d_1}{f_{ETL} + f_{OL} - d_1}, \tag{7}$$

$$d_3 = d_2 + \delta_{ETL+OL}, \tag{8}$$

$$f_2 = \frac{f_{ObL} f_1}{f_1 + f_{ObL} - d_3}, \tag{9}$$

$$\delta_2 = \frac{f_{ObL} d_3}{f_1 + f_{ObL} - d_3}, \tag{10}$$

where $f_{ETL}$ is the focal length of the ETL, $f_{OL}$ is the focal length of the offset lens, $f_{ObL}$ is the focal length of the objective lens, $f_1$ is the compound focal length of the variable focus and offset lenses, $f_2$ is the compound focal length of the ETL, offset lens, and objective lens, $\delta_1$ is the distance from the front principal point of the offset lens to the compound principal point of the variable focus and offset lenses, $\delta_2$ is the distance from the principal point on the front side of the objective lens to the compound principal point of the ETL, offset lens, and objective lens, $d_2$ is the distance from the rear principal point of the objective lens to the front principal point of the offset lens, and $d_3$ is the distance from the rear principal point of the objective lens to the compound principal point of

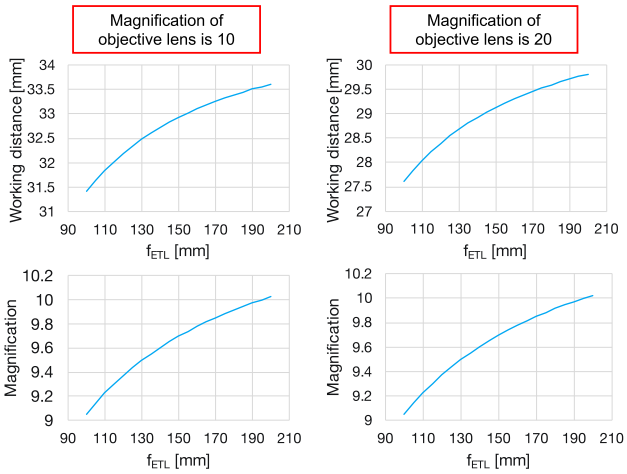



**FIGURE 8.** Relationship between the focal distance, magnification, and WD of the ETL when the magnification of objective lens is 10× and 20×.

the ETL and the offset lens. If the WD is the distance from the objective lens to the in-focus position, the variation of WD, $\Delta WD$ can be expressed using the following equation:

$$\Delta WD = f_2 - \delta_2. \tag{11}$$

By expanding Eq. (11) by $d_1$, the relationship between $d_1$ and $\Delta WD$ can be derived. In addition, by calculating $d_1$, we can derive $f_{ETL}$ from the equation (5). From the above relational expression, the focal length $f_{ETL}$ of the ETL can be adjusted to be the target WD. Also, by using the focal length $f_{IL}$ of the imaging lens and through focal length control, the total magnification $M$ of the system is described as follows:

$$M = \frac{f_{IL}}{f_2}. \tag{12}$$

Changes in WD and $M$ for $f_{ETL}$ in this system are shown in Fig. 8. Figure 8 confirms that the WD of this system can be changed by approximately 2 mm when the magnification of the objective lens is 10×, and by approximately 0.5 mm when the magnification of the objective lens is 20×. Note that the WD mentioned here is derived from the following equation using the WD $WD_{ObL}$ of the objective lens.

$$WD = WD_{ObL} + \Delta WD. \tag{13}$$

### B. EVALUATION OF SETTLING TIME OF ELECTRICALLY TUNABLE LENS

In this subsection, we derive the convergence time of the ETL required to control the focal position. In the evaluation experiment, control commands are given from the control PC, and the ETL is driven by the control commands. Subsequently, the time taken for the focus position to reach the target position and that for the edge intensity to peak is measured. The experimental environment is as shown in Fig. 9. The piezo stage is placed vertically below the mirror, and the USAF 1951 resolution test target is placed on the

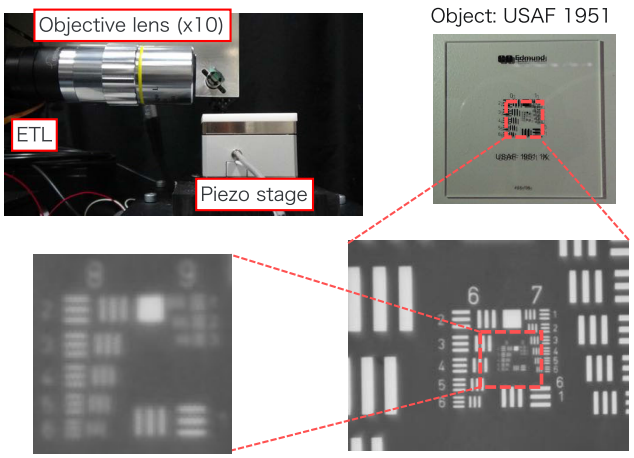



**FIGURE 9.** Experimental setup.

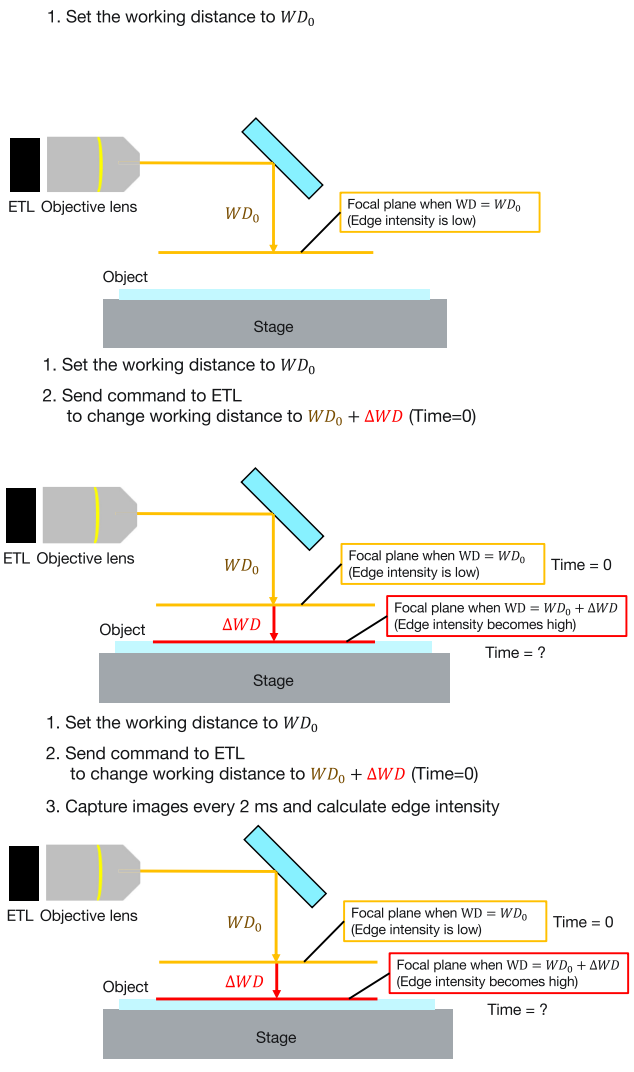



**FIGURE 10.** Evaluation method of ETL response speed.

stage. The evaluation procedure is as follows, and its outline is shown in Fig. 10.

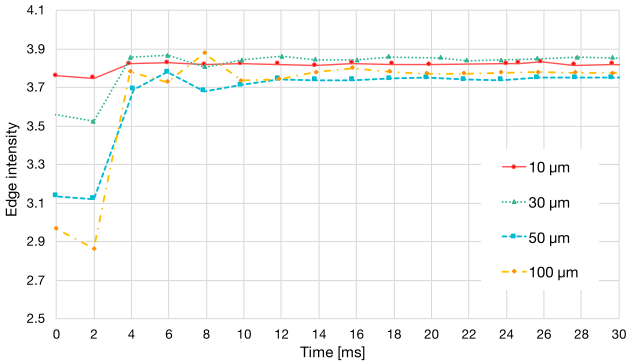



**FIGURE 11.** ETL response speed.

1) Input value of WD $WD_{in}$ is initialized to $WD_0$. Initially, the edge intensity value is low.
2) The input signal is set to $WD_{in} = WD_0 + \Delta WD$ and a command is sent from the control PC. The time $t$ is set to 0.
3) Images are taken every 2 ms and the edge intensity is measured. The edge intensity when the WD becomes equal to the command value is the final edge intensity.

Figure 11 shows the time series data of the edge intensity obtained from the evaluation experiment. The edge intensity when $t$ is 30 ms is set as the final edge intensity. The time when the difference between the final edge intensity and the edge intensity is within 2% is the settling time of the ETL. Figure 11 shows the time series data of edge intensity when the WD is 10, 30, 50, and 100 $\mu$m, respectively. Through the evaluation experiments, the settling times are confirmed to be 4, 8, 8, and 10 ms when the WD of the ETL is 10, 30, 50, and 100 $\mu$m, respectively.

## V. ACQUISITION OF BOTH LARGE FOV AND HIGH-RESOLUTION IMAGES

### A. OVERVIEW OF THE ALGORITHM

During microinjection, the size of the droplet containing the oocytes is approximately 2 to 3 mm. Thus, a wide-range image that shows all the oocytes in the droplet must be captured to avoid the mixing of the oocytes before and after microinjection. Furthermore, images with high-resolution that enable the observation of polar bodies at approximately 20 to 30 fps are also required. Here, we propose an algorithm that presents an expanded large FOV image at a low frame rate and a high-resolution image around the viewpoint selected by the operator at 30 fps. The time required for mirror movement and imaging is 2 ms, and the time required to change the focal length of ETL is 4 ms or more, as shown in IV-B. When acquiring multiple images to expand the FOV, the focal lengths for all viewpoints must be adjusted if multiple images are to be acquired in the tiling order. Therefore, to reduce the time taken to change the WD of the ETL, the viewpoints are scanned in an elliptical trajectory to ensure that the viewpoints with a constant WD can be captured, as shown in Fig. 12. The number of images captured in one

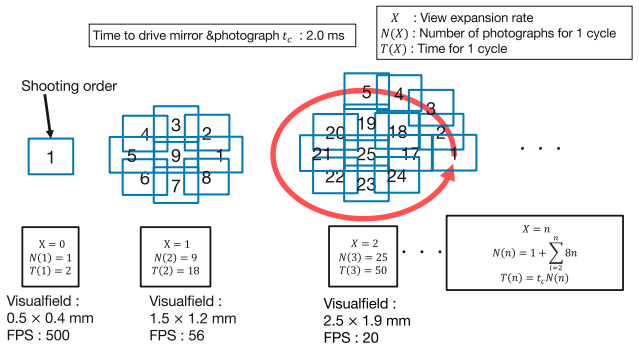



**FIGURE 12.** Order of image acquisition.

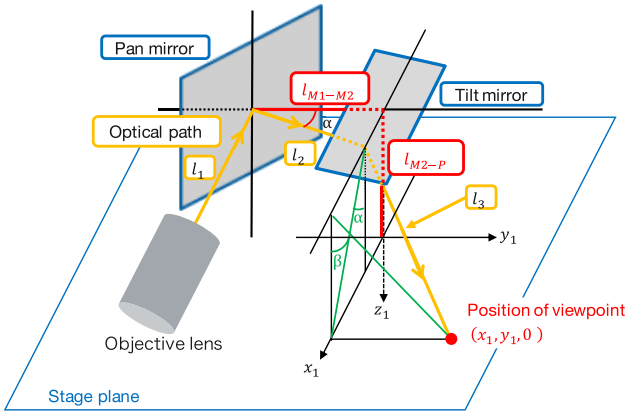



**FIGURE 13.** Optical path between objective lens and stage.

cycle $N(X)$ and the time for one cycle $T(X)$ when the size of FOV is $X$ times are calculated as follows using the time for mirror movement and imaging $t_c$.

$$N(X) = 1 + \sum_{i=2}^{X} 8(X-1), \tag{14}$$

$$T(X) = t_c N(X). \tag{15}$$

Thus, the imaging time per unit area can be reduced to 0.68 of the time required for expanding the FOV by scanning the viewpoints in the tiling sequence.

### B. MIRROR ANGLE DERIVATION FOR LARGE FOV IMAGING

Figure 13 shows the schematic of the optical path from the objective lens to the microscopic stage with two mirrors. $l_1$ is the optical path length from the objective lens to the pan axis mirror, $l_2$ is the optical path length from the pan axis mirror to the tilt axis mirror, $l_3$ is the optical path length from the tilt axis mirror to the stage, $l_{M1-M2}$ is the distance from the pan axis to the tilt axis, $l_{M2-P}$ is the distance from the tilt axis to the microscopic stage, $\alpha$ is the optical angle of the pan axis mirror, and $\beta$ is the optical angle of the tilt axis mirror. The optical path length from the objective lens to the observation object placed on the microscopic stage, i.e., the $WD$ can be

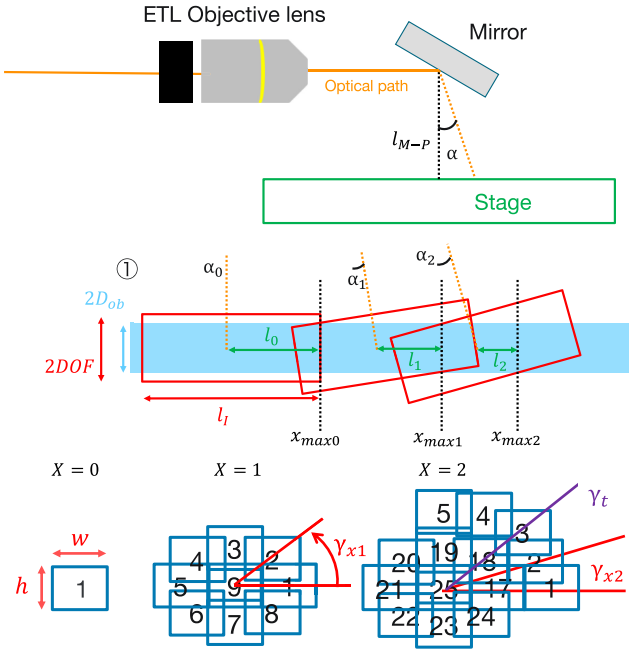



**FIGURE 14.** Derivation of mirror angles for expanded visual field.

derived as follows:

$$WD = l_1 + \frac{l_{M1-M2}}{\cos\alpha} + l_{M2-P}\sqrt{1 + \tan^2\alpha + \tan^2\beta}. \tag{16}$$

Taking the O-$x_1y_1z_1$ coordinate system with the origin at the intersection of the line drawn vertically downward from mirror 2 to the microscopic stage (as shown in Fig. 13), the position of the viewpoint $(x_1, y_1)$ is derived as follows:

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} l_{M1-M2}\tan\alpha + l_{M2-P}\tan\alpha \\ l_{M2-P}\tan\beta \end{pmatrix}. \tag{17}$$

To observe the large FOV plane efficiently, the viewpoints must be selected in a manner that prevents the imaging ranges from overlapping. Fig. 14 shows the schematic of the optical path length from the objective lens to the microscopic stage and the shooting order for deriving viewpoint movement as an ellipse. $l_{M-P}$ is the distance from the mirror to the microscopic stage, $DOF$ is the depth of field, $D_{Ob}$ is the depth of the observation target, and $l_I$ is the length of one image in the $x$-axis direction.

First, we consider the viewpoint of the $y$ direction is 0 ($\beta = 0$), which is the starting point of the elliptical orbit. Let $x_{maxX}$ be the $x$ position of the right edge of an expanded image when the number of scan cycles is $X$, $\alpha_X$ be the angle of the mirror when the number of scan cycles is $X$, $\alpha_{X+1}$ be the mirror angle at the viewpoint of the right edge of the subsequent cycle. $x_{maxX}$, $\alpha_X$, and $\alpha_{X+1}$ are derived as follows:

$$l_X = \min\left\{\frac{1}{2}l_I\cos\alpha_X, DOF\sin\alpha_X + \frac{DOF\cos\alpha_X - D_{ob}}{\tan\alpha_X}\right\} \tag{18}$$

$$x_{maxX} = l_{M-P}\tan\alpha_X + l_X \tag{19}$$

$$\alpha_{X+1} = \tan^{-1}\left(\frac{x_{maxX} + l_X}{l_{M-P}}\right) \tag{20}$$

From the expression (16), the WD with view-expansion rate $X$ is defined as $WD_X$; it is calculated as follows:

$$WD_X = l_1 + \frac{l_{M1-M2}}{\cos\alpha_X} + l_{M2-P}\sqrt{1+\tan^2\alpha_X}. \tag{21}$$

Next, we derive the viewpoints for acquiring images in the elliptical orbit with a WD of $WD_X$, considering the overlap of the imaging areas. The viewpoints for acquiring images are points on the elliptical orbit and are symmetrical vertically and horizontally; thus, the viewpoints in the first quadrant are considered. $\gamma_{nX}$, the angle from the $x$ axis at the $n$th viewpoint position on the elliptical orbit with the view expansion rate $X$ can be expressed as follows:

$$\gamma_{nX} = \begin{cases} (n-1)\tan^{-1}\left(\dfrac{h}{X\cdot w}\right) & (\gamma_{n-1} < \gamma_t) \\ \dfrac{\pi}{2} - (2X+1-n)\tan^{-1}\left(\dfrac{w}{X\cdot h}\right) & (\text{otherwise}), \end{cases}$$
$$(n = 1, 2, \cdots, 2X+1), \tag{22}$$

$$\gamma_t = \frac{w}{h}. \tag{23}$$

From equation (22), the $x_{nX}$ and $y_{nX}$ at the $n$th viewpoint on the elliptical orbit with the view expansion rate $X$ are related as follows:

$$\frac{y_{nX}}{x_{nX}} = \tan\gamma_{nX}. \tag{24}$$

Imaging positions $x_{nX}$ and $y_{nX}$ are derived from Eqs. (21) and (24), and the $x_{nX}$ and $y_{nX}$ are converted to the controlled mirror angles $(\alpha_{nX}, \beta_{nX})$ using Eq. (17).

### C. COMBINED LARGE FOV IMAGE

The captured image, where the mirror angles are respectively $\alpha$ and $\beta$, is tilted by $\alpha$, $\beta$ degrees with respect to the stage plane as shown in Fig. 15. Consider the coordinate system O'-$x_2y_2$ with the image centroid as the origin by taking the $x$ and $y$ axes on the plane tilted by $\alpha$ and $\beta$ degrees. The coordinate system O'-$x_3y_3z_3$ with the image centroid as the origin is defined by taking the $x$ and $y$ axes on the stage plane. To generate a large FOV image by combining the captured images from multiple viewpoints, the coordinate system O'-$x_2y_2$ is converted to the coordinate system O'-$x_3y_3z_3$ as follows:

$$\begin{pmatrix} x_3 \\ y_3 \end{pmatrix} = A\begin{pmatrix} x2 \\ y2 \end{pmatrix},$$
$$A = \begin{pmatrix} \cos\alpha + \sin\alpha\tan\alpha & \sin\beta\tan\alpha \\ \sin\alpha\tan\beta & \cos\beta + \sin\beta\tan\beta \end{pmatrix}. \tag{25}$$

In the O'-$x_2y_2$ coordinate system, the pixel coordinates $(x_p, y_p)$ can be transformed to axis coordinates $(x_2, y_2)$ as follows:

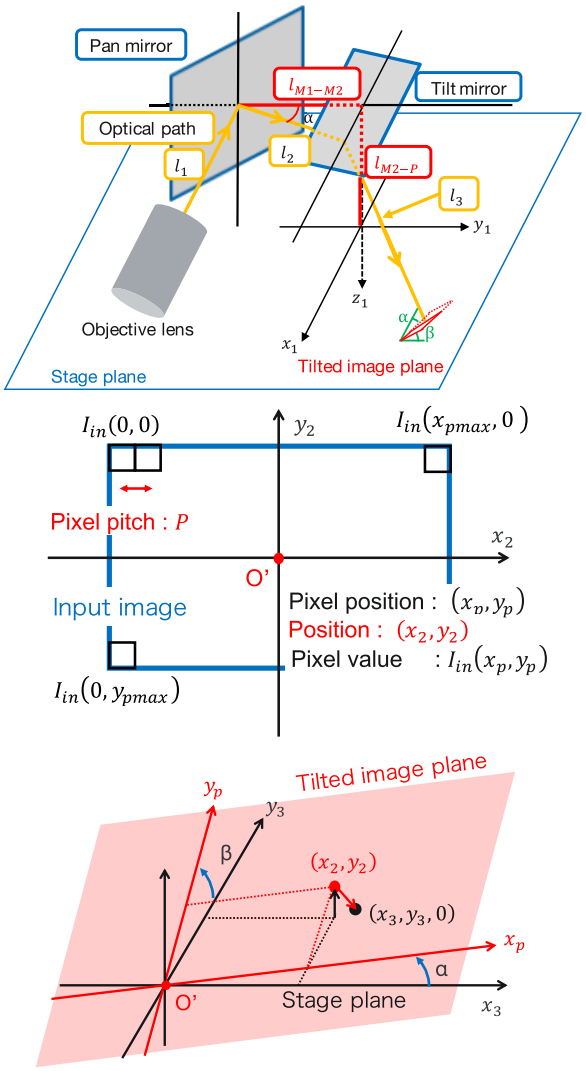



**FIGURE 15.** Conversion of pixel position to xyz coordinates.

$$\begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} \dfrac{(x_p - \frac{x_{pmax}}{2} - 0.5)P}{M} \\ \dfrac{(y_p - \frac{y_{pmax}}{2} - 0.5)P}{M} \end{pmatrix}, \tag{26}$$

where $M$ is the total magnification, $P$ is the pixel pitch, and $(x_{pmax}, y_{pmax})$ is the maximum pixel position.

Finally, by using the imaging position $(x_1, y_1)$ in the O-$x_1y_1z_1$ coordinate system where the mirror angles are $\alpha$ and $\beta$ degrees, the conversion from the pixel coordinates to the world coordinate system O-$x_1y_1z_1$ proceeds as follows:

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_3 \\ y_3 \end{pmatrix},$$
$$= \begin{pmatrix} l_{M1-M2}\tan\alpha + l_{M2-P}\tan\alpha \\ l_{M2-P}\tan\beta \end{pmatrix}$$

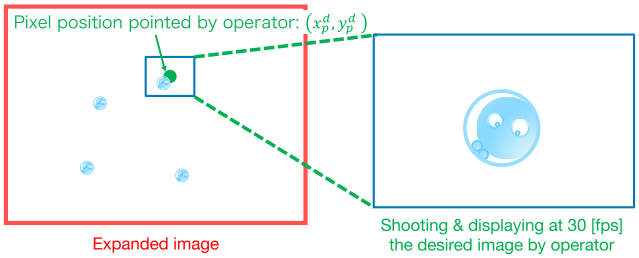



**FIGURE 16.** Acquisition & display of desired image to assist in injection.

$$+A\begin{pmatrix}\dfrac{(x_p-\frac{x_{pmax}}{2}-0.5)P}{M}\\ \dfrac{(y_p-\frac{y_{pmax}}{2}-0.5)P}{M}\end{pmatrix}. \tag{27}$$

The pixel information acquired at each imaging position is combined as pixel information in the world coordinate system to generate a large FOV image using the transformation provided in Eq. (27).

### D. HIGH-RESOLUTION IMAGE ACQUISITION FOR MICROINJECTION

For microinjection, a high-resolution image of the operator's desired viewpoint is presented. The position at which the operator clicks the mouse is set as the desired viewpoint for microinjection, and the images are subsequently acquired and displayed at 40 fps, as shown in Fig. 16. Note that the position at which the mouse is clicked is converted to the mirror angle using Eq. (17) as follows:

$$\begin{pmatrix}\alpha\\ \beta\end{pmatrix}=\begin{pmatrix}\tan^{-1}\dfrac{x_p^d}{l_{M1-M2}+l_{M2-P}}\\ \tan^{-1}\dfrac{y_p^d}{l_{M2-P}}\end{pmatrix}, \tag{28}$$

where $x_p^d$ and $x_p^d$ are the pixel positions in the view-expansion image.

## VI. EVALUATION OF THE PROPOSED SYSTEM

### A. EVALUATION OF RESOLUTION

During microinsemination in the process of cloning or gene manipulation, the nucleus near the polar body of the oocyte is manipulated, and great care should be taken to ensure that the nucleus remains undamaged. Therefore, in general, microinjection requires a resolution that is sufficiently high to identify the polar body, which is approximately 10 $\mu$m in size. In this subsection, the resolution of the proposed system is evaluated using the resolution test target USAF 1951.

The USAF 1951 test target comprises a series of bars of different sizes that are divided into different groups and elements. The smallest group and element that can be imaged clearly by the system is used to calculate the resolution of the system. Fig. 17 shows the images of the test target captured at the center, left, right, top, and bottom of the expanded visual field. Although the resolution is lower in the periphery of the extended visual field, especially in the left and right parts, the

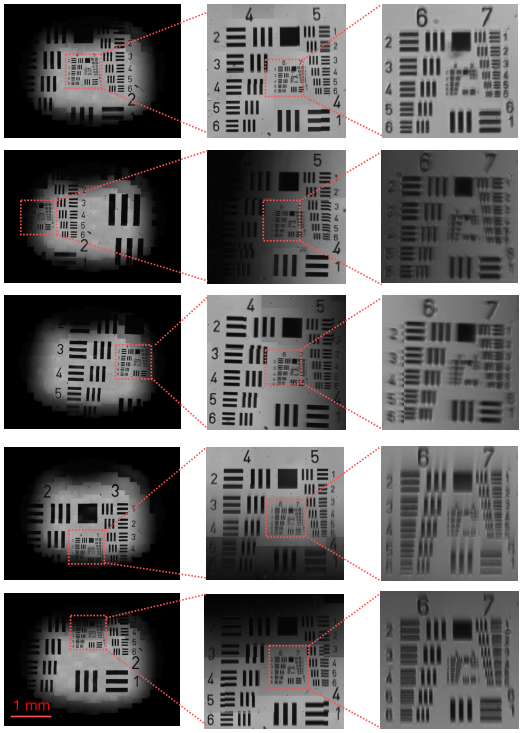



**FIGURE 17.** Expanded images of USAF 1951 test target.

line can be visually confirmed to be approximately 3.1 $\mu$m, which is group number 7 element 3 from Fig. 17. Since the size of the polar body is approximately 10$\mu$m, the resolution of the system is sufficient for microinsemination.

### B. EVALUATION OF TASK TIME IN MICROMANIPULATION

#### 1) EXPERIMENTAL SETUP

Evaluation experiments were conducted to verify that the proposed system improves the task time of microinjection. Referring to the actual microinjection process as shown in Fig. 1, the experimental tasks were performed as follows.

1) Grasp a microbead in workspace1 in the droplet with a pipette guided by the large FOV image.
2) Move the microbead from workspace1 to workspace2.
3) Confirm the operation target with a high-resolution image. Note that, the depth position of the holding pipette is adjusted to focus on the microbead if the acquired image is out-of-focus.
4) Move the microbead from workspace2 to workspace3 guided by the large FOV image.
5) Release the microbead into workspace3, and move the holding pipette back to workspace1.

The experiment was conducted under two different image presentation conditions. The two image presentation conditions included a conventional display that requires changing the magnification of the objective lens, and the newly proposed display that provides both large FOV and high-resolution images. The diameter of the droplet was set as approximately 3.0 mm in both of the conditions. All eight

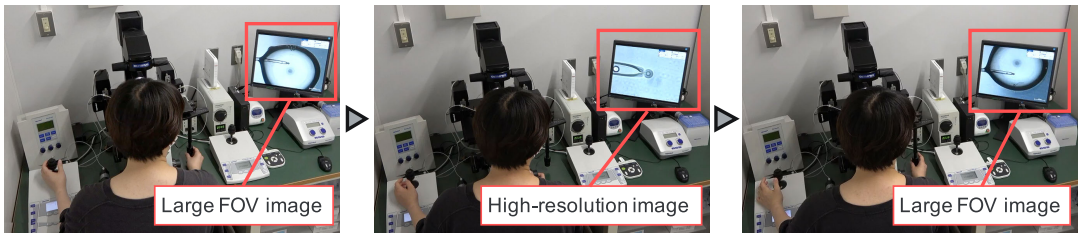



**FIGURE 18.** Experimental scene using a conventional system.

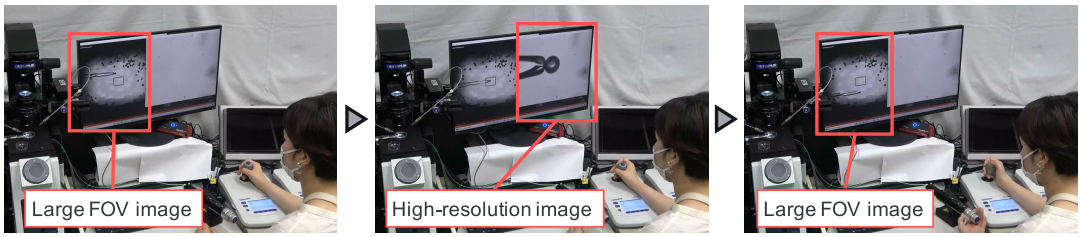



**FIGURE 19.** Experimental scene using the proposed system.

**TABLE 1.** Task time of the experiments.

| | Conventional system | Proposed system |
|---|---|---|
| Subject A | 118 s | 82 s |
| Subject B | 103 s | 68 s |
| Subject C | 89 s | 91 s |
| Subject D | 124 s | 110 s |
| Subject E | 118 s | 94 s |
| Subject F | 168 s | 103 s |
| Subject G | 111 s | 80 s |
| Subject H | 109 s | 72 s |
| Average | 117 s | 87 s |

subjects were inexperienced in micromanipulation and were between 20 and 25 years old. Experimental tasks were conducted after several practice runs through the environments of the two conditions, and each subject performed the experimental tasks four times under each condition. The eight subjects were divided into two groups. The first group performed the experimental tasks using the conventional micromanipulation system, and subsequently using the proposed micromanipulation system. The second group conducted the experiment in the reverse order.

### 2) EXPERIMENTAL RESULTS

Figures 18 and 19 show the experimental scenes using a previously developed conventional micromanipulation system and the proposed micromanipulation system, respectively. The attached video of this paper also shows the experimental scenes. The previous system requires the presented image to be changed from large-FOV to high-resolution, and vice versa. In contrast, the proposed system presents both the large-FOV and high-resolution images simultaneously; that is, the presented image does not need to be changed. Table 1 shows the mean task times of each condition, and Fig. 20 shows the box plot of the experimental results. The average

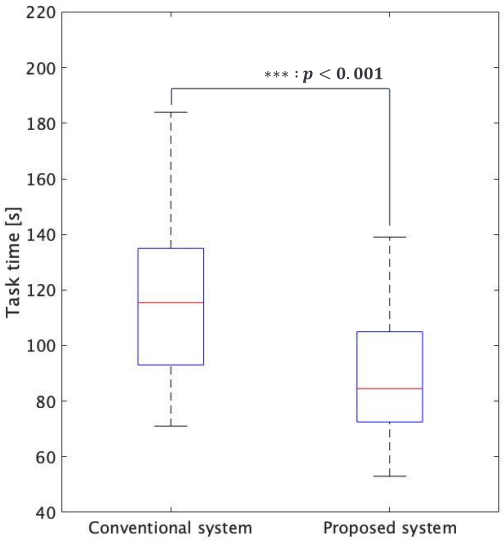



**FIGURE 20.** Box plot of the experimental results.

task time using the conventional and proposed systems were 117.4 s and 87.3 s, respectively. The proposed system significantly improved the task time by approximately 26 % ($p < 0.001$). These results indicate that our micromanipulation system streamlines the task of micromanipulation by eliminating the need to change the objective lens.

## VII. CONCLUSION

In this study, we developed a novel view-expansion microscope system with transmitted illumination that can image the target objects placed on a plane stage. The view-expansion microscope system uses the focal position adjustment function of an ETL to prevent defocusing caused by the change of

the optical path length. Then, we proposed a micromanipulation system that enables both large FOV and high-resolution imaging with free viewpoint selection. It, consequently, reduces the workload of the operator and simplifies the process of microinjection. Algorithms to acquire and present both large FOV and high-resolution images were implemented in the proposed system for efficient microinjection. The effectiveness of the proposed system was verified by evaluating the resolution of images presented to the operator and through experiments on moving microbeads performed by inexperienced subjects. The acquisition and display of depth information in the microinjection process will be undertaken in future work.

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